CS-2005 – Database Systems – Spring 2026

Assignment # 03

Maximum Marks: 10

Post Date: April 28, 2026

Due Date: May 09, 2026

**Note:**

- Plagiarism of any kind will not be tolerated.
- You need to submit/ upload the SCAN copy of your hand-written assignment on google
- classroom.
- Late submissions will not be accepted.

## Question 01:

In 2026, due to the rapid growth of online education platforms, the Global E-Learning Monitoring System (GEMS) has developed a centralized database to track courses, enrollments, and performance.
The GEMS database includes the following relations:

| Relation | Attributes | Description |
|---|---|---|
| PLATFORM | (Platform_ID, Platform_Name, Country, Rating) | Stores details of e-learning platforms |
| COURSE | (Course_ID, Course_Name, Category, Difficulty_Level) | Stores courses offered on platforms |
| ENROLLMENT | (Enroll_ID, Course_ID, Platform_ID, Enroll_Date, Completion_Rate) | Tracks student enrollments and completion |
| INSTRUCTOR | (Instructor_ID, Platform_ID, Instructor_Name, Expertise) | Stores instructors working on platforms |
| COURSE_ACTIVITY | (Enroll_ID, Instructor_ID, Activity_Type, Status) | Records instructor activities and course completion status |

You are a Database Analyst working for GEMS. Write the Relational Algebra expression and one line explanation of each command for the following:

1. List names of platforms offering advanced-level courses (Difficulty_Level = "Advanced") in 2026
2. Retrieve instructor names who handled Data Science courses successfully completed (Status = "Completed")
3. Find platforms that have no instructors registered
4. Display courses that are offered on more than one platform

5. List all AI-related courses (Category = "AI") with platform names and completion rate > 85

## Question 02:

Consider a database with the following schema:

Library (libID, libName, city, headID)
Staff (staffID, name, address, salary, libID)
Member (memberID, name, city, phone)
Borrow (borrowID, memberID, libID, date, fine)
Book (bookID, title, genre, price, isAvailable)
BorrowDetails (borrowID, bookID, quantity)
Publisher (publisherID, name, city, contact)
Publishes (publisherID, bookID, publishDate, cost)

Describe the relations that would be produced by the following relational algebra operations (Textual meaning required) and also provide sql queries of each algebraic expression:

1. $\pi$_title ($\sigma$_isAvailable = 0 $\wedge$ price > 1000 (Book))
2. $\pi$_title ($\sigma$_(genre = 'Fiction' $\wedge$ price > 800) $\vee$ genre = 'History' (Book))
3. $\pi$_title, price ($\sigma$_genre = 'Science' $\wedge$ isAvailable = 1 (Book))
4. $\pi$_title, price/10 ($\sigma$_genre = 'Science' $\wedge$ isAvailable = 1 (Book))
5. $\pi$_publisherID (Publishes $\bowtie$ Book $\bowtie$ $\sigma$_title = 'AI Basics' (Book))
6. $\pi$_Staff.name, Library.libName ($\sigma$_Staff.libID = Library.libID $\wedge$ Staff.address = Library.city (Staff $\bowtie$ Library))
7. $\pi$_staffID ($\sigma$_salary $\neq$ 60000 (Staff))
8. $\pi$_publisherID ($\sigma$_city = 'Karachi' (Publisher)) $\cap$ $\pi$_publisherID ($\sigma$_city = 'Islamabad' (Publisher))
9. $\pi$_title (Book) − $\pi$_title (Book $\bowtie$ Publishes)

## Question 03

Two transactions on a bank system are given**:**

| T1 | T2 |
|---|---|
| read_item(A); | read_item(A); |
| A := A − N; | A := A + M; |
| write_item(A); | write_item(A); |

## Tasks:

a. Write a non-serial schedule
b. Construct precedence graph (step-by-step)
c. Check if conflict-serializable
d. If not, write correct schedule
e. Real-world impact?

## Question 04

Consider the four transactions T1, T2, T3 and T4 and the schedules S1 and S2 given below.

**Transactions:**

T1: r1(x); w1(x); r1(y)
T2: r2(y); w2(y)
T3: r3(x); w3(x)
T4: r4(y); w4(y)

**Schedules**

S1:
r1(x), r2(y), r3(x), r4(y), w1(x), w2(y), w3(x), w4(y)
S2:
r3(x), w3(x), r1(x), r1(y), w1(x), r2(y), w2(y), r4(y), w4(y)

### Tasks:

1. Draw precedence graphs
2. Check serializability
3. Write equivalent serial order

## Question 05

Consider Table: Account(Account_ID, Balance) whose Initial Balance = 5000
Transactions:

- **T1:** Transfer 1000 (subtract)
- **T2:** Add bonus 2000

The execution interleaves as follows:

1. T1 reads 5000
2. T2 reads 5000
3. T1 → 4000
4. T2 → 7000
5. T1 writes 4000
6. T2 writes 7000

Tasks:

1. Identify problem (e.g., lost update)
2. Final balance after execution
3. Correct balance if executed serially

## Question 06 (a)

Explain the timestamp ordering protocol for concurrency control by thinking of transactions as people given a number when they arrive, so everyone should act in order. What problems can occur in this basic system, such as transactions getting rejected or restarted again and again? Also, describe how strict timestamp ordering improves the situation by keeping things more controlled and avoiding messy conflicts, and support your explanation with a simple example.

## Question 06 (b):

Suppose two transactions are running at the same time:

T1 moves $300 from Account A to B

T2 moves $200 from Account B to A.
Explain how, under concurrency control, a deadlock situation can arise when each transaction holds one account and waits for the other. Then describe a suitable method to either prevent this situation or resolve it once it occurs.

THE END!